TidyTuesday Week 45

Pranay Gundam

Wednesday 4th November, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we throw two unrelated economic time series into a blender and solemnly report whatever sludge comes out. This week the random number generator, in its infinite wisdom, decided to introduce goods-producing workers in Springfield, Massachusetts to a nursing home price index, bank returns in Missouri to IT orders, and residential electric power prices to America's spending on burgers and beer. A truly curated salon of strangers who have never met.

Monday kicked things off with a bang: Springfield's goods-producing employment versus proprietary and government nursing home prices posted a raw R-squared of 0.78, which, if you squint and ignore everything you know, looks downright impressive. Then we detrended it, and the relationship evaporated so completely that the R-squared slumped to 0.02 and the p-value climbed to a majestic 0.999 – as close to "absolutely nothing" as statistics will politely allow. In other words, factory jobs in Springfield were quietly shrinking while nursing home prices were quietly rising, and our regression mistook two people walking in opposite directions for a passionate romance. Classic trend mirage, and exactly the kind of thing this project exists to point and laugh at.

Wednesday gave us IT manufacturing orders versus the return on equity for Missouri banks, a series so beloved it was literally discontinued. The raw version was significant, which is unsurprising, but here's the mildly interesting part: the relationship actually survived detrending, holding on with a p-value around 0.01. It's weak – an R-squared of 0.06 is not going to get anyone tenure – but it technically clung to life after we stripped out the shared drift, which puts it in the rarer "huh, maybe there's a faint something there" category rather than the usual "pure coincidence" bin. I wouldn't bet my retirement on Missouri banking returns forecasting the nation's IT appetite, but I'll grudgingly tip my hat.

Then Thursday served up the week's grand finale: residential electric power prices in the East North Central census division versus national spending on food services and drinking places. The raw fit was a glorious 0.74, and even after detrending it held up at an R-squared of 0.21 with a p-value small enough to make a p-value enthusiast weep. Does your bar tab rise with your electric bill? Possibly it's inflation gently inflating everything at once, or possibly electricity and eating out are both just things people do more of as life gets pricier – but this one at least refused to die when we took away its trend crutch. As always, the mandatory disclaimer: these are randomly paired series, nobody is claiming anyone causes anyone, and spurious correlation remains the house specialty. If anything surprised us this week, it's that two of our three blind

dates actually exchanged phone numbers.

2 Date: 2026-11-02

Series ID: SMU25781000600000001
This series is titled All Employees: Goods Producing in Springfield, MA-CT (NECTA) and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1990-01-01 and the observation end date is 2024-12-01.The popularity of this series is 1.

Series ID: PRGONUPIHCSA
This series is titled Medical Services Expenditures by Provider: Nursing Homes: Proprietary and Government Nursing Homes Price Index and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-01-01 and the observation end date is 2021-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_PRGONUPIHCSA	**R-squared:**	0.783
Model:	OLS	**Adj. R-squared:**	0.772
Method:	Least Squares	**F-statistic:**	72.01
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	4.64e-08
Time:	21:06:51	**Log-Likelihood:**	-73.979
No. Observations:	22	**AIC:**	152.0
Df Residuals:	20	**BIC:**	154.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	186.2043	11.787	15.797	0.000	161.616	210.793
value_fred_SMU25781000600000001	-2.2652	0.267	-8.486	0.000	-2.822	-1.708

Omnibus:	5.822	**Durbin-Watson:**	0.251
Prob(Omnibus):	0.054	**Jarque-Bera (JB):**	4.103
Skew:	1.043	**Prob(JB):**	0.129
Kurtosis:	3.353	**Cond. No.**	333.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

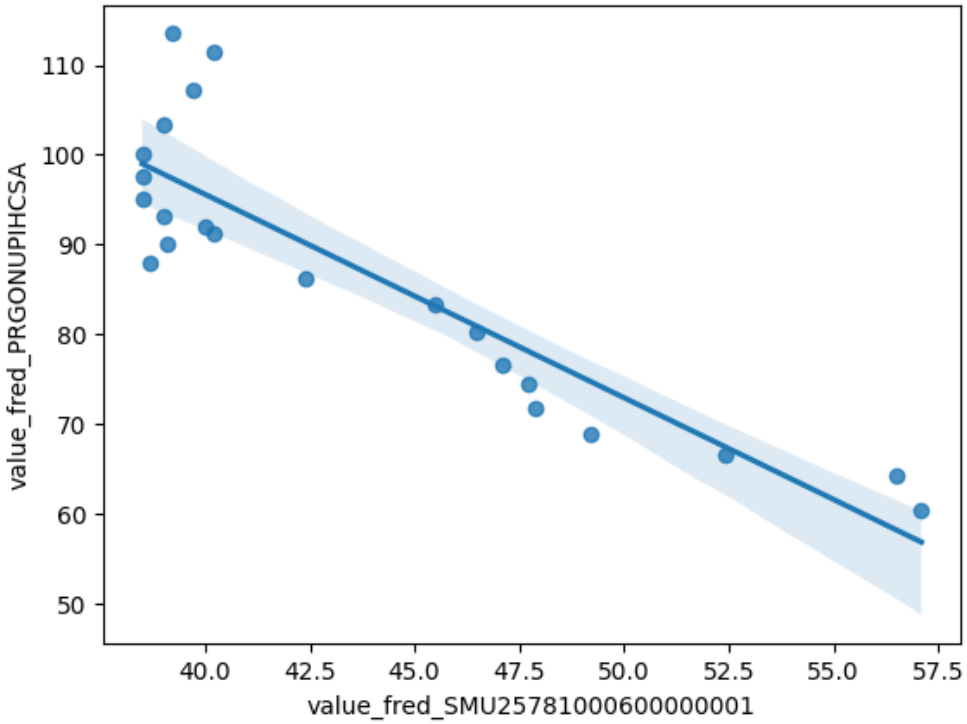


Figure 1: Raw Regression Plot for 2026-11-02

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_PRGONUPIHCSA_detrended	**R-squared:**	0.021
Model:	OLS	**Adj. R-squared:**	-0.028
Method:	Least Squares	**F-statistic:**	0.4223
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	0.523
Time:	21:06:51	**Log-Likelihood:**	-42.096
No. Observations:	22	**AIC:**	88.19
Df Residuals:	20	**BIC:**	90.37
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.027e-13	0.367	1.1e-12	1.000	-0.765	0.765
value_fred_SMU25781000600000001_detrended	-0.0866	0.133	-0.650	0.523	-0.365	0.191

Omnibus:	2.262	**Durbin-Watson:**	0.331
Prob(Omnibus):	0.323	**Jarque-Bera (JB):**	1.135
Skew:	-0.097	**Prob(JB):**	0.567
Kurtosis:	1.904	**Cond. No.**	2.75

Notes:

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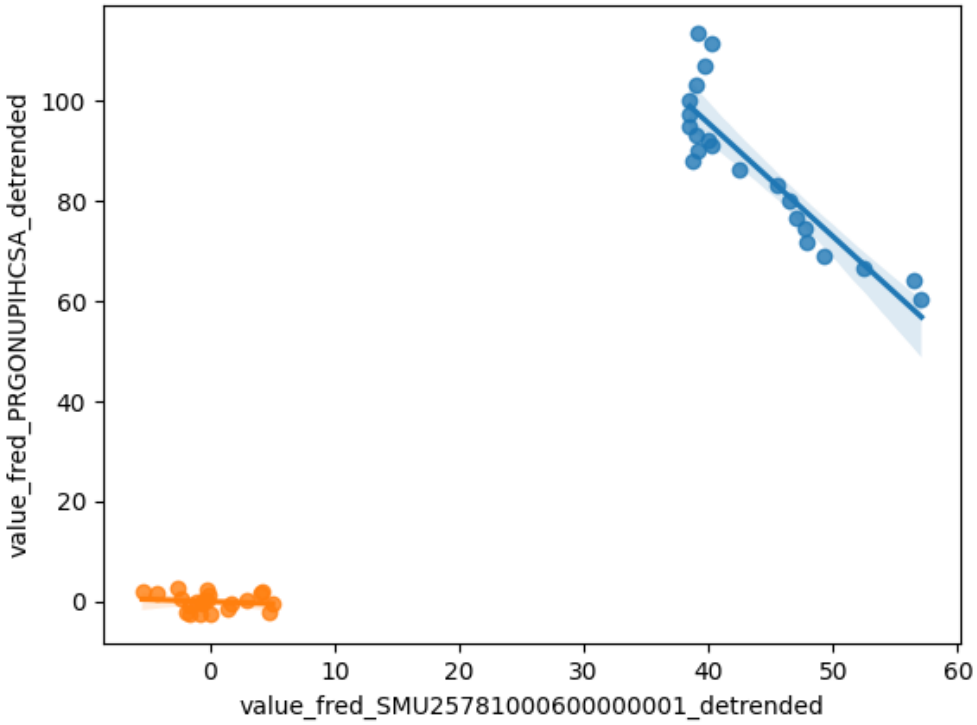


Figure 2: Detrended Regression Plot for 2026-11-02

3 Date: 2026-11-03

Series ID: AITINO
This series is titled Manufacturers' New Orders: Information Technology Industries and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted.The observation start date is 1992-02-01 and the observation end date is 2026-09-01.The popularity of this series is 14.

Series ID: MOROE
This series is titled Return on Average Equity for Banks in Missouri (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1984-01-01 and the observation end date is 2020-07-01.The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MOROE	**R-squared:**	0.135
Model:	OLS	**Adj. R-squared:**	0.128
Method:	Least Squares	**F-statistic:**	17.51
Date:	Tue, 03 Nov 2026	**Prob (F-statistic):**	5.69e-05
Time:	18:56:19	**Log-Likelihood:**	-307.15
No. Observations:	114	**AIC:**	618.3
Df Residuals:	112	**BIC:**	623.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.9234	2.332	0.825	0.411	-2.697	6.544
value_fred_AITINO	0.0004	9.45e-05	4.185	0.000	0.000	0.001

Omnibus:	41.602	**Durbin-Watson:**	0.161
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	88.948
Skew:	-1.485	**Prob(JB):**	4.84e-20
Kurtosis:	6.147	**Cond. No.**	1.70e+05

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.7e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

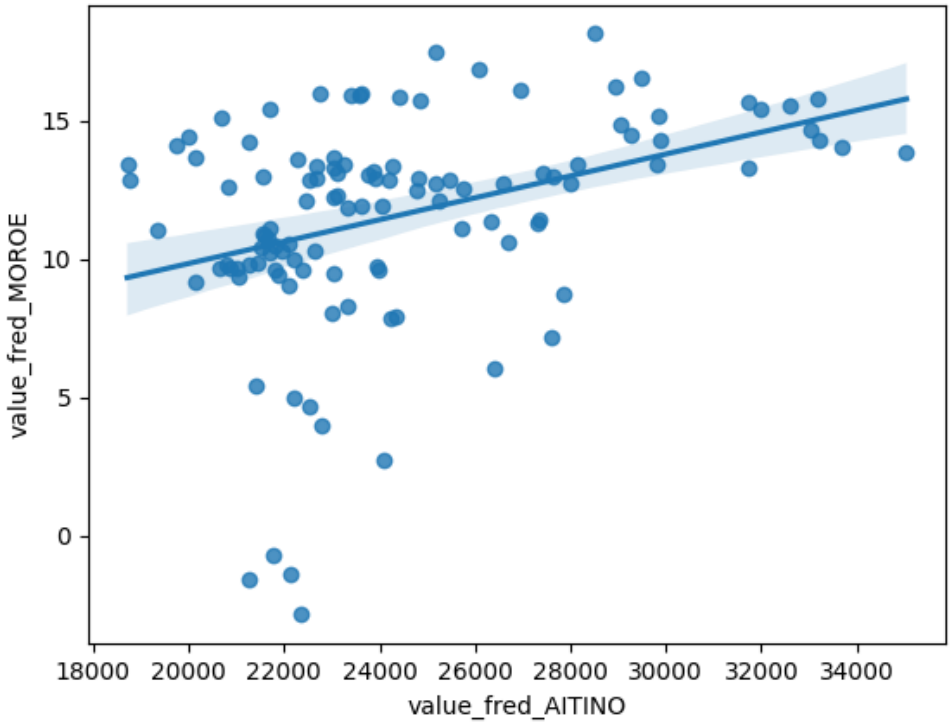


Figure 3: Raw Regression Plot for 2026-11-03

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MOROE_detrended	**R-squared:**	0.057
Model:	OLS	**Adj. R-squared:**	0.049
Method:	Least Squares	**F-statistic:**	6.779
Date:	Tue, 03 Nov 2026	**Prob (F-statistic):**	0.0105
Time:	18:56:19	**Log-Likelihood:**	-295.86
No. Observations:	114	**AIC:**	595.7
Df Residuals:	112	**BIC:**	601.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-7.674e-13	0.306	-2.5e-12	1.000	-0.607	0.607
value_fred_AITINO_detrended	0.0002	9.13e-05	2.604	0.010	5.68e-05	0.000

Omnibus:	60.278	**Durbin-Watson:**	0.190
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	192.530
Skew:	-1.974	**Prob(JB):**	1.56e-42
Kurtosis:	7.994	**Cond. No.**	3.35e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.35e+03. This might indicate that there are strong multicollinearity or other numerical problems.

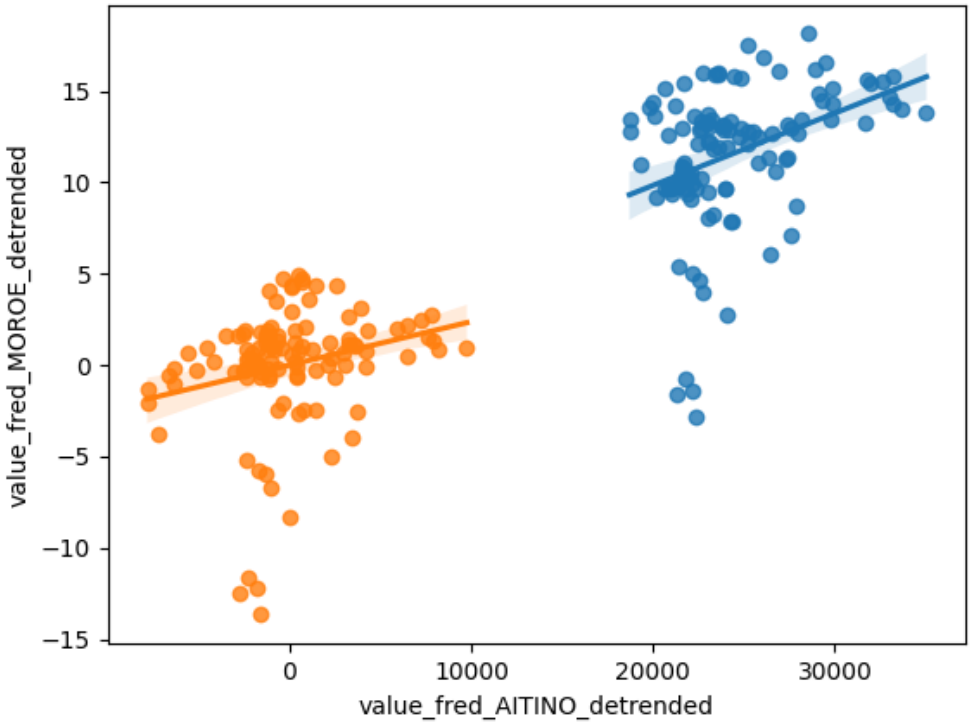


Figure 4: Detrended Regression Plot for 2026-11-03

4 Date: 2026-11-04

Series ID: WPU054121013
This series is titled Producer Price Index by Commodity: Fuels and Related Products and Power: Residential Electric Power for East North Central Census Division and has a frequency of Monthly. The units are Index Dec 2007=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2007-12-01 and the observation end date is 2026-09-01.The popularity of this series is 1.

Series ID: MRTSSM722USN
This series is titled Retail Sales: Food Services and Drinking Places and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1992-01-01 and the observation end date is 2026-08-01.The popularity of this series is 19.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MRTSSM722USN	**R-squared:**	0.737
Model:	OLS	**Adj. R-squared:**	0.734
Method:	Least Squares	**F-statistic:**	318.7
Date:	Wed, 04 Nov 2026	**Prob (F-statistic):**	8.31e-35
Time:	22:07:01	**Log-Likelihood:**	-1192.6
No. Observations:	116	**AIC:**	2389.
Df Residuals:	114	**BIC:**	2395.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2.164e+04	5164.973	-4.190	0.000	-3.19e+04	-1.14e+04
value_fred_WPU054121013	575.5062	32.238	17.852	0.000	511.643	639.370

Omnibus:	8.963	**Durbin-Watson:**	0.814
Prob(Omnibus):	0.011	**Jarque-Bera (JB):**	10.211
Skew:	-0.481	**Prob(JB):**	0.00606
Kurtosis:	4.089	**Cond. No.**	1.25e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

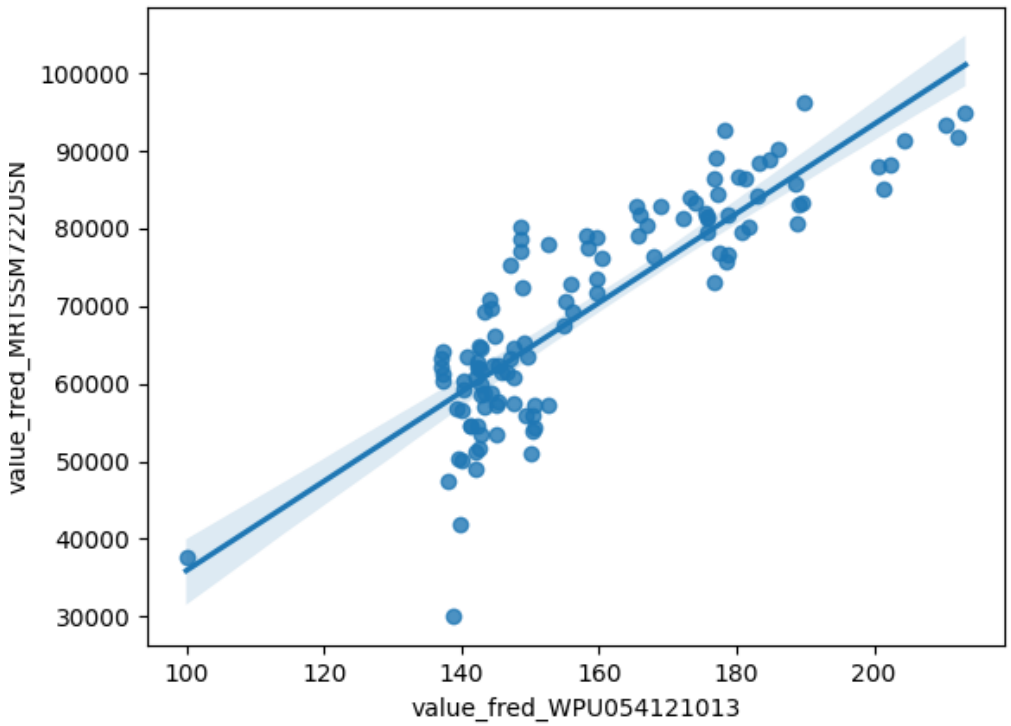


Figure 5: Raw Regression Plot for 2026-11-04

[2] The condition number is large, 1.25e+03. This might indicate that there are strong multicollinearity or other numerical problems.

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MRTSSM722USN_detrended	**R-squared:**	0.211
Model:	OLS	**Adj. R-squared:**	0.204
Method:	Least Squares	**F-statistic:**	30.56
Date:	Wed, 04 Nov 2026	**Prob (F-statistic):**	2.08e-07
Time:	22:07:01	**Log-Likelihood:**	-1180.4
No. Observations:	116	**AIC:**	2365.
Df Residuals:	114	**BIC:**	2370.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.527e-09	595.106	2.57e-12	1.000	-1178.899	1178.899
value_fred_WPU054121013_detrended	318.6043	57.635	5.528	0.000	204.429	432.779

Omnibus:	41.308	**Durbin-Watson:**	0.835
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	110.393
Skew:	-1.327	**Prob(JB):**	1.07e-24
Kurtosis:	6.975	**Cond. No.**	10.3

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

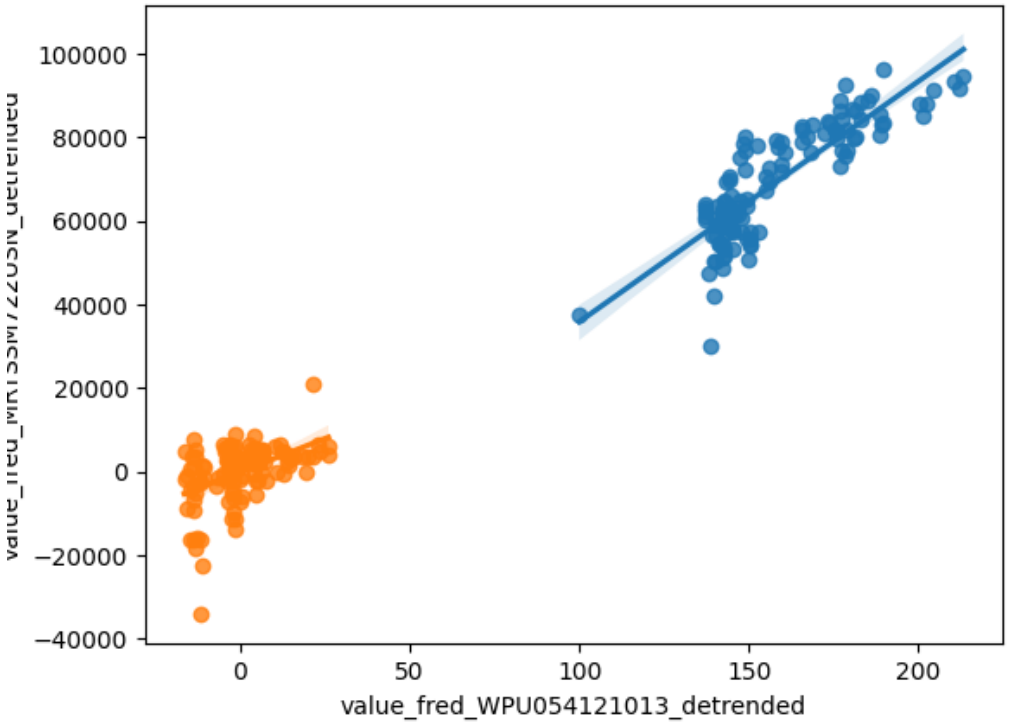


Figure 6: Detrended Regression Plot for 2026-11-04